6.3 Scalar Product and Vector Product

Next we recall the scalar product and vector product of two vectors as follows.

Definition 6.1

Given two vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ the scalar product (or dot product) is denoted by $\vec{a} \cdot \vec{b}$ and is calculated by

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3,$$

and the vector product (or cross product) is denoted by $\vec{a} \times \vec{b}$, and is calculated by

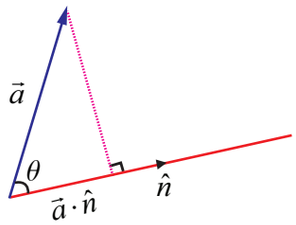
$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Note

$\vec{a} \cdot \vec{b}$ is a scalar, and $\vec{a} \times \vec{b}$ is a vector.

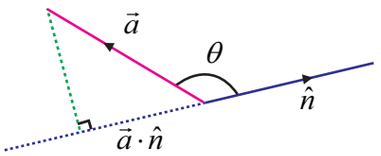
6.3.1 Geometrical interpretation

Geometrically, if $\vec{a}$ is an arbitrary vector and $\hat{n}$ is a unit vector, then $\vec{a} \cdot \hat{n}$ is the projection of the vector $\vec{a}$ on the straight line on which $\hat{n}$ lies. The quantity $\vec{a} \cdot \hat{n}$ is positive if the angle between $\vec{a}$ and $\hat{n}$ is acute, see Fig. 6.4 and negative if the angle between $\vec{a}$ and $\hat{n}$ is obtuse see Fig. 6.5.



Positive dot product

Fig. 6.4



Negative dot product

Fig. 6.5

If $\vec{a}$ and $\vec{b}$ are arbitrary non-zero vectors, then $|\vec{a} \cdot \vec{b}| = \left| |\vec{b}| \vec{a} \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right) \right| = \left| |\vec{a}| \vec{b} \cdot \left(\frac{\vec{a}}{|\vec{a}|} \right) \right|$ and so $|\vec{a} \cdot \vec{b}|$ means either the length of the straight line segment obtained by projecting the vector $|\vec{b}|\vec{a}$ along the direction of $\vec{b}$ or the length of the line segment obtained by projecting the vector $|\vec{a}|\vec{b}$ along the direction of $\vec{a}$. We recall that $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$, where θ is the angle between the two vectors $\vec{a}$ and $\vec{b}$. We recall that the angle between $\vec{a}$ and $\vec{b}$ is defined as the measure from $\vec{a}$ to $\vec{b}$ in the counter clockwise direction.

The vector $\vec{a} \times \vec{b}$ is either $\vec{0}$ or a vector perpendicular to the plane parallel to both $\vec{a}$ and $\vec{b}$ having magnitude as the area of the parallelogram formed by coterminus vectors parallel to $\vec{a}$ and $\vec{b}$. If $\vec{a}$ and $\vec{b}$ are non-zero vectors, then the magnitude of $\vec{a} \times \vec{b}$ can be calculated by the formula

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}||\sin\theta|, \text{ where } \theta \text{ is the angle between } \vec{a} \text{ and } \vec{b}.$$

Two vectors are said to be coterminus if they have same initial point.

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